Hong Kiat, Tan

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Education

University of California, Los Angeles

- **Ph.D.** Candidate, Mathematics. Aug 2021 – End 2026
- Advisor: Andrea L. Bertozzi. (GPA: 3.98/4.00)
- **Master of Arts** in Mathematics. Aug 2021 – Jun 2022

National University of Singapore

- **Bachelor of Science (Honours)** in Applied Mathematics with Highest Distinction. Aug 2017 – May 2021
- Second Major in Physics and Minor in Statistics. (GPA: 4.97/5.00)
- *Ho Family Prize* – Top graduating student in Applied Mathematics, with 28 A+'s in Math/Physics/Statistics courses.

Industrial/Work Experiences

Applied Scientist Intern, Amazon Search — June 2026 –

- Working on deep learning/LLM-based approaches for improving search relevance for natural language queries.

Quantitative Research Intern, WorldQuant Intraday Team — Sep 2025 – Dec 2025

- Developed tensorized singular value decomposition infrastructure in Python for statistical analysis.
- Productionized features from limit order book data pipeline via SLURM-orchestrated C++ infrastructure.

Data Scientist Intern, Amazon Search Data Science and Economics — Jun 2025 – Sep 2025

- Pioneered novel framework orchestrating multi-agent LLMs with ℓ^0-changepoint detection for 78 search metrics in 17 locales to generate interpretable economic insights.
- Wrote a paper as a first author that is accepted at ICLR Time Series in Age of Large Models (TSALM) Workshop 2026.
- Built agentic production pipeline on AWS Bedrock AgentCore using Strands/LangChain with ECS/Docker orchestration.
- Spearheaded agentic coding initiatives and authored internal documentation on MCP servers and data science workflows.

Academic Experiences

Research Fellow, Supervised Program for Alignment Research — Feb 2026 – Present

- Building a multi-agent framework leveraging attribution graphs and transcoders for automated circuit analysis of LLM internals for real-time internal monitoring, extending Neuronpedia's circuit tracer library.

Graduate Research Assistant, UCLA — 2022 – Present

- Developed a modified vector autoregressive framework using double machine learning and causal inference for confounding-adjusted lag detection in time series, utilizing EconML, TabPFN, and PyTorch in Python. (Submitted.)
- Architected an object-oriented Python framework for traffic optimization on road networks, integrating game-theoretic models and PDE solvers on directed graphs, validated on Lahaina's road topology. (arXiv:2603.29055, 2026.)
- Designed numerical PDE schemes in Python with penalized regression for physics-informed flux functions and proved convergence by developing and applying novel concepts in differential topology. (SIAM J. Math. Anal., 2026.)

Graduate Teaching Assistant, UCLA — 2021 – 2025

- Developed 786 pages of instructional materials across 10 quarters for advanced mathematics courses (Algorithms, Probability, Graduate PDEs, Mathematical Finance, and Analysis), with an average teaching evaluation of 8.6/9.0.

Undergraduate Research Assistant, NUS — 2020 – 2021

- Developed a numerical scheme with hypothesis testing and regression in R for quantum field theory. (Phys. Rev. D, 2022.)
- Co-authored a 148-page paper on a fundamental conjecture in mathematical general relativity. (arXiv:2402.16250, 2024.)

Undergraduate Research Assistant, UNC – Chapel Hill — 2019

- Designed Bayesian hierarchical models for astrophysical data with MCMC samplers in R. (Astrophys. J., 2020.)

Undergraduate Teaching Assistant, NUS — 2019 – 2021

- Served as TA for discrete structures and Python programming across 5 semesters, with average feedback score of 4.8/5.0.

Technical Skills & Professional Activities

- *Programming Languages*: Python, C++ (Intermediate), PostgreSQL, R, LaTeX.
- *Agentic Tooling*: Claude Code, Codex, OpenClaw, Skills, MCPs, LangChain, Anthropic/OpenAI/Claude Code SDKs.
- *Cloud & DevOps*: AWS (AgentCore, Lambda, ECS/ECR, Bedrock), GCP, Modal, RunPod, Docker, Linux (SLURM), Git.
- *Libraries/Frameworks*: PyTorch, vLLM, Hugging Face, Strands, wandb, EconML, TabPFN, scikit-learn, pandas.
- *Coursework*: Statistical Learning, High-dim Statistics, Optimization, Causal Inference, Functional Analysis, PDEs.
- *Service*: Reviewer for AISTATS 2026, UAI 2026, NeurIPS 2026, ICML Workshop 2026.
- *Certification:* BlueDot Technical AI Safety.